Text Models for Measuring Ideology

POS6933: Computational Social Science

Jake S. Truscott, Ph.D

University of Florida

Spring 2026

Overview

- Housekeeping
- Class 9 Problem Set Review
- `Wordscores`
- `Wordfish`
- `Wordshoal`
- Office Hours (Until 6:00)

Housekeeping

- **Reminders**:
  - Slides due to me and peer reviewers by 11:59pm on Sunday (Schedule *Next*)
  - Final materials (Paper, Appendix, and Final Slides) due 4/24 (Canvas)
  - Rubric & help document located on Canvas (*Files*) – Be sure to follow formatting requirements...
- I will stay until 6:00 today if you want to chat about your papers

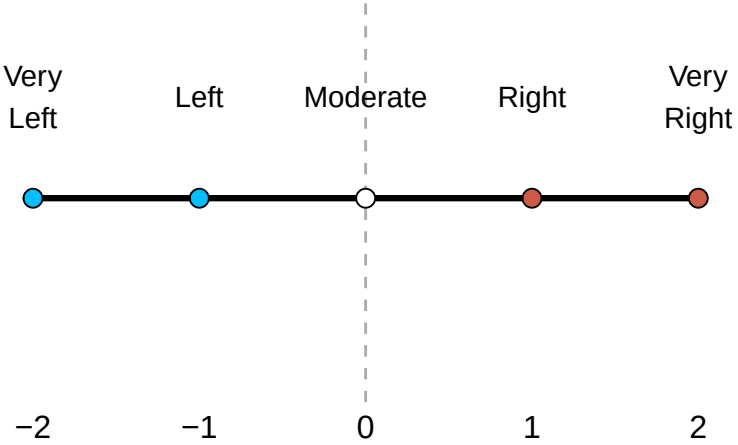
Housekeeping – Presentation Order

Day	Student	Reviewer 1	Reviewer 2
4/12	Arora, Manya	Mamun	Revilla, Valeria
4/12	Cui, Lixiao	Meneley, Cory	Sarver, Sarah
4/12	Giron-Khalil, Jailyn	Mohammed, Suren	Sultana, Abida
4/12	Hao, Tianshu	Oneil, Connor	Tierman, Hunter
4/12	Mamun	Sarver, Sarah	Wang, Qianxi
4/12	Meneley, Cory	Sultana, Abida	Arora, Manya
4/12	Mohammed, Suren	Tierman, Hunter	Cui, Lixiao
4/19	Islam, Md Sahidul	Revilla, Valeria	Trenchi, Alejandro
4/19	Oneil, Connor	Trenchi, Alejandro	Giron-Khalil, Jailyn
4/19	Revilla, Valeria	Wang, Qianxi	Hao, Tianshu
4/19	Sarver, Sarah	Arora, Manya	Islam, Md Sahidul
4/19	Sultana, Abida	Cui, Lixiao	Mamun
4/19	Tierman, Hunter	Giron-Khalil, Jailyn	Meneley, Cory
4/19	Trenchi, Alejandro	Hao, Tianshu	Mohammed, Suren
4/19	Wang, Qianxi	Islam, Md Sahidul	Oneil, Connor

Class 9 Problem Set Review

- All did very well – Nice!

Scaling Ideology



Scaling Ideology (Cont.)

- **How I Was Taught to Think of It**: Rather than using right-hand side of the equation to solve for the left, flip the intuition – Use DV to estimate optimal IV(s).

Scaling Ideology (Cont.)

- **How I Was Taught to Think of It**: Rather than using right-hand side of the equation to solve for the left, flip the intuition – Use DV to estimate optimal IV(s).
- Assumptions:
  1. **Stable Latent Preferences**: Assumes actors have stable, coherent, underlying ideological positions that guide their behavior.

Scaling Ideology (Cont.)

- **How I Was Taught to Think of It**: Rather than using right-hand side of the equation to solve for the left, flip the intuition – Use DV to estimate optimal IV(s).
- Assumptions:
  1. **Stable Latent Preferences**: Assumes actors have stable, coherent, underlying ideological positions that guide their behavior.
  2. **Behavior Reflects Ideology**: Treats votes or language as meaningful signals of those latent preferences (not pure noise or strategy).

Scaling Ideology (Cont.)

- **How I Was Taught to Think of It**: Rather than using right-hand side of the equation to solve for the left, flip the intuition – Use DV to estimate optimal IV(s).
- Assumptions:
  1. **Stable Latent Preferences**: Assumes actors have stable, coherent, underlying ideological positions that guide their behavior.
  2. **Behavior Reflects Ideology**: Treats votes or language as meaningful signals of those latent preferences (not pure noise or strategy).
  3. **Common Dimensional Structure**: Assumes actors operate in a shared ideological space (often low-dimensional) that is comparable across observations.

Scaling Ideology (Goals)

- **Recover Latent Position(s)**: Using observable behaviors as signals (ex: votes on legislation), estimate an underlying ideological dimension that isn't directly observed but we know exists and can be represented given both directionality (e.g., *Left-Right*) and magnitude (e.g., *Very Left-Very Right*).

Scaling Ideology (Goals)

- **Recover Latent Position(s)**: Using observable behaviors as signals (ex: votes on legislation), estimate an underlying ideological dimension that isn't directly observed but we know exists and can be represented given both directionality (e.g., *Left-Right*) and magnitude (e.g., *Very Left-Very Right*).
- **Place Actors (Groups) On a Spectrum**: Relative location of actors (groups) in a common space denotes cardinality (and proximity) of relationship in that dimension.

Text Models for Ideology (Benefits)

- Binary votes sometimes insufficient to fully understand elite political behaviors
- Text leverages high dimensional structure – more variance and perhaps better explanatory power.
- Text captures nuance in reasoning, framing, and emphasis that votes alone cannot reveal
- Enables dynamic and context-sensitive measurement of ideology across issues, time, and audiences

Models Today

- Going to focus on 3 models: `Wordscores`, `Wordfish`, and `Wordshoal`
- **Important**: View subsequent models as methodological improvements on preceding

Wordscores (Laver, Benoit, and Garry)

- Developed by Laver, Benoit, and Garry (2003) – Probably not(?) first text-based model of ideology, but certainly first to gain significant use in social sciences.
- It's effectively (sort of) Naive Bayes + Value Projection onto a Reference Scale + Rescaling

Wordscores (Laver, Benoit, and Garry)

- Developed by Laver, Benoit, and Garry (2003) – Probably not(?) first text-based model of ideology, but certainly first to gain significant use in social sciences.
- It's effectively (sort of) Naive Bayes + Value Projection onto a Reference Scale + Rescaling
- *Recall...* for standard Naive Bayes, we:
  1. Estimate how likely each word w is to appear in each class k: $p(w \mid k)$

Wordscores (Laver, Benoit, and Garry)

- Developed by Laver, Benoit, and Garry (2003) – Probably not(?) first text-based model of ideology, but certainly first to gain significant use in social sciences.
- It's effectively (sort of) Naive Bayes + Value Projection onto a Reference Scale + Rescaling
- *Recall...* for standard Naive Bayes, we:
  1. Estimate how likely each word w is to appear in each class k: $p(w \mid k)$
  2. For new document D_i, we use those learned word probabilities to calculate π_{ik} (probability that D_i belongs in each possible class: $p(k \mid D_i)$

Wordscores (Laver, Benoit, and Garry)

- Developed by Laver, Benoit, and Garry (2003) – Probably not(?) first text-based model of ideology, but certainly first to gain significant use in social sciences.
- It's effectively (sort of) Naive Bayes + Value Projection onto a Reference Scale + Rescaling
- *Recall...* for standard Naive Bayes, we:
  1. Estimate how likely each word w is to appear in each class k: $p(w \mid k)$
  2. For new document D_i, we use those learned word probabilities to calculate π_{ik} (probability that D_i belongs in each possible class: $p(k \mid D_i)$
  3. Classification assigned to the document because it has the highest probability

Wordscores (Laver, Benoit, and Garry) – Cont.

- **What `Wordscores` does instead**: Estimate how likely it is that each word belongs to a reference document: $p(w \mid r)$, where A_r represents the ideological score of those reference documents.

Wordscores (Laver, Benoit, and Garry) – Cont.

- **What `Wordscores` does instead**: Estimate how likely it is that each word belongs to a reference document: $p(w \mid r)$, where A_r represents the ideological score of those reference documents.
- $S_w = \sum_r p(r \mid w) A_r$ – the score of some word S_w is the sum of the probabilities it belongs to each reference document times the score associated with those documents.

Wordscores (Laver, Benoit, and Garry) – Cont.

- **What `Wordscores` does instead**: Estimate how likely it is that each word belongs to a reference document: $p(w \mid r)$, where A_r represents the ideological score of those reference documents.
- $S_w = \sum_r p(r \mid w) A_r$ – the score of some word S_w is the sum of the probabilities it belongs to each reference document times the score associated with those documents.
- New documents v are finally rescaled to account for weighted averages of word scores – such that a document's position is the average ideology of the words it uses (from the reference documents) and weighted by the word frequences:

$$S_v = \sum_w p(w \mid v) S_w$$

Wordscores Example

Word	Left Ref.	Right Ref.	TOTAL
welfare	8	1	9
tax	2	6	8
military	1	7	8
TOTAL	11	14	25

- 3 Words (*welfare*, *tax*, *military*)
- 2 reference documents:

$$A_{Left} = (-1)$$
$$A_{Right} = (+1)$$

Wordscores Example (Cont.)

Word	Left Ref.	Right Ref.	TOTAL
welfare	8	1	9
tax	2	6	8
military	1	7	8
TOTAL	11	14	25

- Let's start with word probabilities:

  $p(\text{left} \mid \text{welfare}) = \frac{8}{9} = 0.889$

  $p(\text{right} \mid \text{welfare}) = \frac{1}{9} = 0.111$

Wordscores Example (Cont.)

Word	Left Ref.	Right Ref.	TOTAL
welfare	8	1	9
tax	2	6	8
military	1	7	8
TOTAL	11	14	25

- Let's start with word probabilities:

  $p(\text{left} \mid \text{welfare}) = \frac{8}{9} = 0.889$

  $p(\text{right} \mid \text{welfare}) = \frac{1}{9} = 0.111$

  $p(\text{left} \mid \text{tax}) =$

Wordscores Example (Cont.)

Word	Left Ref.	Right Ref.	TOTAL
welfare	8	1	9
tax	2	6	8
military	1	7	8
TOTAL	11	14	25

- Let's start with word probabilities:

$p(\text{left} \mid \text{welfare}) = \frac{8}{9} = 0.889$

$p(\text{right} \mid \text{welfare}) = \frac{1}{9} = 0.111$

$p(\text{left} \mid \text{tax}) = \frac{2}{8} = 0.25$

$p(\text{right} \mid \text{tax}) =$

Wordscores Example (Cont.)

Word	Left Ref.	Right Ref.	TOTAL
welfare	8	1	9
tax	2	6	8
military	1	7	8
TOTAL	11	14	25

- Let's start with word probabilities:

$p(\text{left} \mid \text{welfare}) = \frac{8}{9} = 0.889$

$p(\text{right} \mid \text{welfare}) = \frac{1}{9} = 0.111$

$p(\text{left} \mid \text{tax}) = \frac{2}{8} = 0.25$

$p(\text{right} \mid \text{tax}) = \frac{2}{8} = 0.75$

$p(\text{left} \mid \text{military}) =$

Wordscores Example (Cont.)

Word	Left Ref.	Right Ref.	TOTAL
welfare	8	1	9
tax	2	6	8
military	1	7	8
TOTAL	11	14	25

- Let's start with word probabilities:

$p(\text{left} \mid \text{welfare}) = \frac{8}{9} = 0.889$

$p(\text{right} \mid \text{welfare}) = \frac{1}{9} = 0.111$

$p(\text{left} \mid \text{tax}) = \frac{2}{8} = 0.25$

$p(\text{right} \mid \text{tax}) = \frac{2}{8} = 0.75$

$p(\text{left} \mid \text{military}) = \frac{1}{8} = 0.125$

$p(\text{right} \mid \text{military}) =$

Wordscores Example (Cont.)

Word	Left Ref.	Right Ref.	TOTAL
welfare	8	1	9
tax	2	6	8
military	1	7	8
TOTAL	11	14	25

- Let's start with word probabilities:

$$p(\text{left} \mid \text{welfare}) = \frac{8}{9} = 0.889$$
$$p(\text{right} \mid \text{welfare}) = \frac{1}{9} = 0.111$$
$$p(\text{left} \mid \text{tax}) = \frac{2}{8} = 0.25$$
$$p(\text{right} \mid \text{tax}) = \frac{2}{8} = 0.75$$
$$p(\text{left} \mid \text{military}) = \frac{1}{8} = 0.125$$
$$p(\text{right} \mid \text{military}) = \frac{7}{8} = 0.875$$

Wordscores Example (Cont.)

Word	$p(\text{left} \mid w)$	$p(\text{right} \mid w)$
welfare	0.889	0.111
tax	0.25	0.75
military	0.125	0.875

- Now we can compute the word scores using $S_w = \sum_r p(r \mid w) A_r$:

$$S_{\text{welfare}} = 0.889(-1_{Neg}) + 0.111(1_{Pos}) = -0.778$$

$$S_{\text{tax}}$$

Wordscores Example (Cont.)

Word	$p(\text{left} \mid w)$	$p(\text{right} \mid w)$
welfare	0.889	0.111
tax	0.25	0.75
military	0.125	0.875

- Now we can compute the word scores using $S_w = \sum_r p(r \mid w) A_r$:

$$S_{\text{welfare}} = 0.889(-1_{Neg}) + 0.111(1_{Pos}) = -0.778$$

$$S_{\text{tax}} = 0.25(-1_{Neg}) + 0.75(1_{Pos}) = 0.50$$

$$S_{\text{military}}$$

Wordscores Example (Cont.)

Word	$p(\text{left} \mid w)$	$p(\text{right} \mid w)$
welfare	0.889	0.111
tax	0.25	0.75
military	0.125	0.875

- Now we can compute the word scores using $S_w = \sum_r p(r \mid w) A_r$:

$$S_{\text{welfare}} = 0.889(-1_{Neg}) + 0.111(1_{Pos}) = -0.778$$

$$S_{\text{tax}} = 0.25(-1_{Neg}) + 0.75(1_{Pos}) = 0.50$$

$$S_{\text{military}} = 0.125(-1_{Neg}) + 0.875(1_{Pos}) = 0.75$$

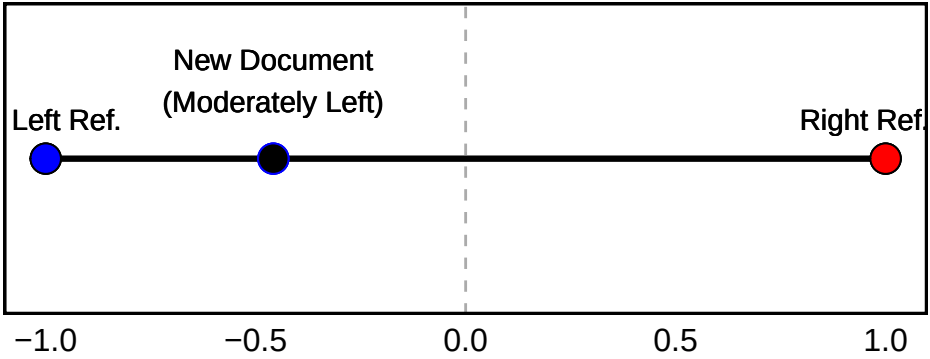
Wordscores Example (Cont.)

- We can now start computing document scores for new documents $S_v = \sum_w p(w \mid v) S_w$
- Suppose a new document (S_v) contains the following: **Welfare** (3), **Tax** (1), and **Military** (0).
- We know that $p(\text{welfare} \mid v) = \frac{3}{4} = 0.75$, and $p(\text{tax} \mid v) = \frac{1}{4} = 0.25$.
- Solve for S_v

Wordscores Example (Cont.)

- We can now start computing document scores for new documents $S_v = \sum_w p(w \mid v) S_w$
- Suppose a new document (S_v) contains the following: **Welfare** (3), **Tax** (1), and **Military** (0).
- We know that $p(\text{welfare} \mid v) = \frac{3}{4} = 0.75$, and $p(\text{tax} \mid v) = \frac{1}{4} = 0.25$.
- Solve for $S_v = 0.75(-0.778) + 0.25(0.5) = (-0.458)$

Wordscores Example (Cont.)



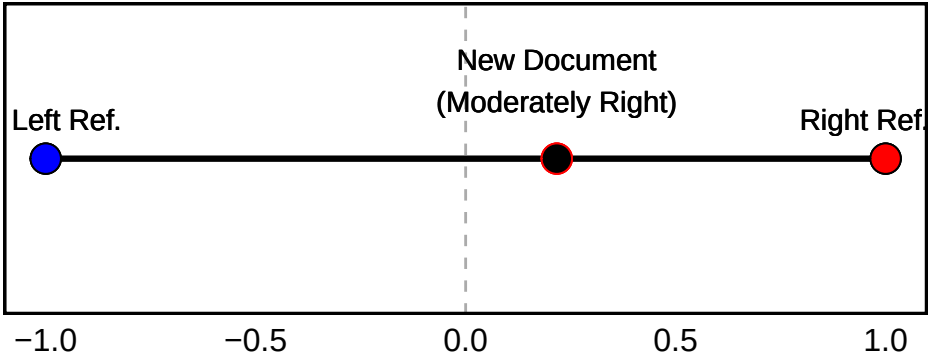
Wordscores Example (Cont.)

- Recall: $S_v = \sum_w p(w \mid v) S_w$
- Suppose another new document (S_v) contains the following: **Welfare** (2), **Tax** (5), and **Military** (3).
- $p(\text{Welfare} \mid v) = 0.20$, $p(\text{Tax} \mid v) = 0.5$, and $p(\text{Military} \mid v) = 0.33$
- Solve for S_v

Wordscores Example (Cont.)

- Recall: $S_v = \sum_w p(w \mid v) S_w$
- Suppose another new document (S_v) contains the following: **Welfare** (2), **Tax** (5), and **Military** (3).
- $p(\text{Welfare} \mid v) = 0.20$, $p(\text{Tax} \mid v) = 0.5$, and $p(\text{Military} \mid v) = 0.33$
- Solve for $S_v = 0.20(-0.778) + 0.5(0.25) + 0.33(0.75) = 0.217$

Wordscores Example (Cont.)



Wordfish (Slapin and Proksch)

- `Wordfish` is a poisson factor model that (like Wordscores) tries to scale the ideological positions of documents.

Wordfish (Slapin and Proksch)

- `Wordfish` is a poisson factor model that (like Wordscores) tries to scale the ideological positions of documents.
- It does **not** require reference documents – it is **unsupervised**
- Uses maximum likelihood to infer both a document's latent position θ_i and word discrimination β_j (how strongly a word separates documents)

Wordfish Specification

$$x_{ij} \sim \text{Poisson}(\lambda_{ij})$$

$$\lambda_{ij} = \exp(\alpha_j + \psi_j + \beta_j \theta_i)$$

Where:

- x_{ij} = the count of word j in document i
- α_i = Document-specific verbosity (value to control for length of document – ensures θ_i reflects content differences, not just document size)
- ψ_j = Word-specific baseline frequency
- β_j = Word-specific discrimination
- θ_i = Latent position of document i (**What we're most interested in**!)

Wordfish Example (Note)

- *Note*: Model uses MLE – iterative fitting of parameters to maximize a likelihood function
- WE can simplify a lot for small examples like this to demonstrate intuition of model – We'll substitute values for α_i, θ_i, ψ_j, and β_j
- We'll assume values greater than zero discriminate right-leaning

Wordfish Example

Document	welfare	tax	military	TOTAL
Doc 1	8	2	1	11
Doc 2	1	6	7	14
Doc 3	3	1	0	4

Parameter	Doc 1	Doc 2	Doc 3
α_i	2.4	2.7	1.2
θ_i	0.5	1.0	-0.3

Word	ψ_j	β_j
welfare	0.2	0.8
tax	0.1	0.5
military	0.3	1.2

Wordfish Example (Cont.)

Compute the expected counts: $\lambda_{ij} = \exp(\alpha_i + \psi_j + \beta_j \theta_i)$:

Word	Document	Formula	Expected Count (λ_{ij})
welfare	Doc 1	$\exp(2.4_{\alpha_i} + 0.2_{\psi_j} + 0.8_{\beta_j} \cdot 0.5_{\theta_i})$	$\exp(3.0) \approx 20.1$
tax	Doc 1	$\exp(2.4 + 0.1 + 0.5 \times 0.5)$	$\exp(2.75) \approx 15.6$
military	Doc 1	$\exp(2.4 + 0.3 + 1.2 \times 0.5)$	$\exp(3.3) \approx 27.1$

Wordfish Example (Cont.)

Compute the expected counts: $\lambda_{ij} = \exp(\alpha_i + \psi_j + \beta_j \theta_i)$:

Word	Document	Formula	Expected Count (λ_{ij})
welfare	Doc 1	$\exp(2.4_{\alpha_i} + 0.2_{\psi_j} + 0.8_{\beta_j} \cdot 0.5_{\theta_i})$	$\exp(3.0) \approx 20.1$
tax	Doc 1	$\exp(2.4 + 0.1 + 0.5 \times 0.5)$	$\exp(2.75) \approx 15.6$
military	Doc 1	$\exp(2.4 + 0.3 + 1.2 \times 0.5)$	$\exp(3.3) \approx 27.1$
welfare	Doc 2	$\exp(2.7 + 0.2 + 0.8 \times 1)$	$\exp(3.7) \approx 40.4$
tax	Doc 2	$\exp(2.7 + 0.1 + 0.5 \times 1)$	$\exp(3.3) \approx 27.1$
military	Doc 2	$\exp(2.7 + 0.3 + 1.2 \times 1)$	$\exp(4.2) \approx 66.7$

Wordfish Example (Cont.)

Compute the expected counts: $\lambda_{ij} = \exp(\alpha_i + \psi_j + \beta_j\theta_i)$:

Word	Document	Formula	Expected Count (λ_{ij})
welfare	Doc 1	$\exp(2.4_{\alpha_i} + 0.2_{\psi_j} + 0.8_{\beta_j} \cdot 0.5_{\theta_i})$	$\exp(3.0) \approx 20.1$
tax	Doc 1	$\exp(2.4 + 0.1 + 0.5 \times 0.5)$	$\exp(2.75) \approx 15.6$
military	Doc 1	$\exp(2.4 + 0.3 + 1.2 \times 0.5)$	$\exp(3.3) \approx 27.1$
welfare	Doc 2	$\exp(2.7 + 0.2 + 0.8 \times 1)$	$\exp(3.7) \approx 40.4$
tax	Doc 2	$\exp(2.7 + 0.1 + 0.5 \times 1)$	$\exp(3.3) \approx 27.1$
military	Doc 2	$\exp(2.7 + 0.3 + 1.2 \times 1)$	$\exp(4.2) \approx 66.7$
welfare	Doc 3	$\exp(1.2 + 0.2 + 0.8 \times -0.3)$	$\exp(1.16) \approx 3.1$
tax	Doc 3	$\exp(1.2 + 0.1 + 0.5 \times -0.3)$	$\exp(1.15) \approx 3.1$
military	Doc 3	$\exp(1.2 + 0.3 + 1.2 \times -0.3)$	$\exp(1.14) \approx 3.1$

Wordfish Example – Key Takeaways

- Doc 2 is the most right leaning ($\theta_i = 1$), while Doc 1 is moderately right and Doc 3 is moderately left.
- Doc 3 is the shortest document ($\alpha_i = 1.2$), while Doc 2 is the longest – Doc 1 appears to be almost in between.
- Military proves to be the most important discriminating word ($\beta_j = 1.2$), showing strong preference as a word that's important for identifying right-leaning documents – e.g., Doc 1 and 2, where $x_{Doc2,\text{Military}} = 66.7$ and $x_{Doc1,\text{Military}} = 27.1$, versus $x_{Doc3,\text{Military}} = 3.1$

Wordfish R Manual Example

```r
docs <- c('Doc1', 'Doc2', 'Doc3') # Documents
words <- c('welfare', 'tax', 'military') # Discriminating Words

alpha_i <- c(Doc1 = 2.4, Doc2 = 2.7, Doc3 = 1.2) # Verbosity
theta_i <- c(Doc1 = 0.5, Doc2 = 1.0, Doc3 = -0.3)  # Latent Ideology
psi_j <- c(welfare = 0.2, tax = 0.1, military = 0.3)  # Baseline Word Freq.
beta_j <- c(welfare = 0.8, tax = 0.5, military = 1.2) # Word Discrimination

lambda <- matrix(0, nrow = length(docs), ncol = length(words),
                 dimnames = list(docs, words)) # Matrix to Input Recovered Lambda_ij Values

for (d in docs) {
  for (w in words) {
    lambda[d, w] <- exp(alpha_i[d] + psi_j[w] + beta_j[w] * theta_i[d])
  }
} # Function -- For Each Doc(i)-Word(j) Pair, Recover Lambda_ij
```

Wordfish R Manual Example (Cont.)

```r
as.data.frame(lambda) %>%
  mutate(document = rownames(lambda)) %>%
  tidyr::pivot_longer(cols = -document,
                      names_to = "word",
                      values_to = "lambda") %>%
  mutate(lambda = round(lambda, 2))
```

```
# A tibble: 9 x 3
  document word     lambda
  <chr>    <chr>     <dbl>
1 Doc1     welfare   20.1
2 Doc1     tax       15.6
3 Doc1     military  27.1
4 Doc2     welfare   40.4
5 Doc2     tax       27.1
6 Doc2     military  66.7
7 Doc3     welfare    3.19
8 Doc3     tax        3.16
9 Doc3     military   3.13
```

Wordfish R Quanteda Example – Irish Dail

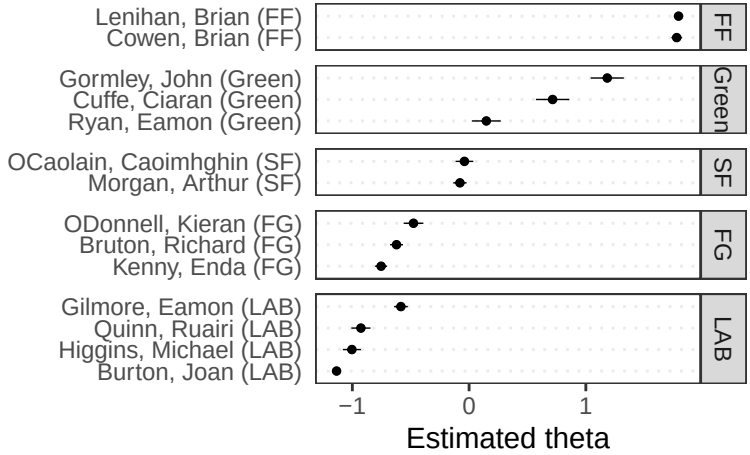
```
dail <- quanteda::tokens(quanteda.textmodels::data_corpus_irishbudget2010, remove_punct = TRUE)
dail_dfm <- quanteda::dfm(dail) # DFM of Tokenized Dail Corpus

quanteda::docnames(dail_dfm) # Legislator Names
```

```
 [1] "Lenihan, Brian (FF)"       "Bruton, Richard (FG)"      "Burton, Joan (LAB)"        "Morgan, Arthur
 [5] "Cowen, Brian (FF)"         "Kenny, Enda (FG)"          "ODonnell, Kieran (FG)"     "Gilmore, Eamon
 [9] "Higgins, Michael (LAB)"    "Quinn, Ruairi (LAB)"       "Gormley, John (Green)"     "Ryan, Eamon (Gr
[13] "Cuffe, Ciaran (Green)"     "OCaolain, Caoimhghin (SF)"
```

```
dail_wordfish <- quanteda.textmodels::textmodel_wordfish(dail_dfm,
                                    dir = c(6, 5)) # Specifying Kenny on Left & Cowen on Right
```

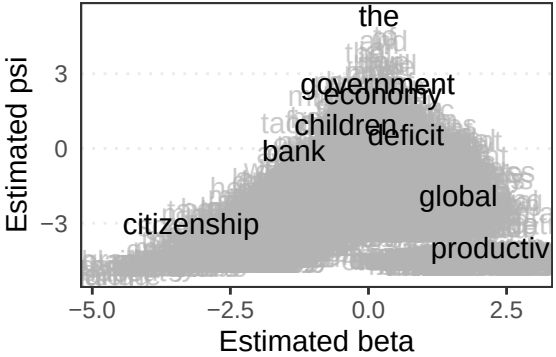
Wordfish R Quanteda Example – Irish Dail (Cont.)

```
quanteda.textplots::textplot_scale1d(dail_wordfish, groups = dail_dfm$party) # 1D Plot by Party
```



Wordfish R Quanteda Example – Irish Dail (Cont.)

```
quanteda.textplots::textplot_scale1d(dail_wordfish, margin = "features",  highlighted = c("government", "
```



Wordshoal (Lauderdale and Herzog)

- `Wordshoal` can be seen as a direct improvement on `Wordfish`, such that it provides for the recovery of legislator-specific θ_i pooled across several debates.

Wordshoal (Lauderdale and Herzog)

- `Wordshoal` can be seen as a direct improvement on `Wordfish`, such that it provides for the recovery of legislator-specific θ_i pooled across several debates.
- Rather than using speeches from a single debate or issue, we can use a (psuedo) or fully Bayesian process to recover a shared latent θ_i.

Wordshoal (Lauderdale and Herzog)

- `Wordshoal` can be seen as a direct improvement on `Wordfish`, such that it provides for the recovery of legislator-specific θ_i pooled across several debates.
- Rather than using speeches from a single debate or issue, we can use a (psuedo) or fully Bayesian process to recover a shared latent θ_i.
- We could do much of the same work to recover λ_{ij} for each author and then subsequently weight these values by the discrimination parameter β_j.
- We won't... (at least not by hand)

Wordshoal R Manual Example

```
docs <- data.frame(document = c('Doc1', 'Doc2', 'Doc3', 'Doc4'),
                   author   = c(1, 1, 2, 2),
                   alpha    = c(2.0, 1.8, 2.2, 2.0)) # Docs - theta_i & alpha_i

words <- data.frame(word = c('welfare', 'tax', 'military'),
                    psi  = c(0.2, 0.1, 0.3),
                    beta = c(0.8, 0.5, 1.2)) # Words -- Psi_j & beta_j

theta <- data.frame(author = c(1, 2),
                    theta  = c(-1, 1)) # Author Ideologies


wordshoal_grid <- expand.grid(document = docs$document,
                              word = words$word) %>%
  left_join(docs, by="document") %>%
  left_join(words, by="word") %>%
  left_join(theta, by="author") %>% # All Document-Word Combos
  mutate(lambda = exp(alpha + psi + beta*theta),
         lambda = round(lambda, 2)) # Calculate lambda_ij
```

Wordshoal R Manual Example (Cont.)

```r
head(tibble(wordshoal_grid))
```

```
# A tibble: 6 x 8
  document word    author alpha   psi  beta theta lambda
  <chr>    <chr>    <dbl> <dbl> <dbl> <dbl> <dbl>  <dbl>
1 Doc1     welfare      1   2     0.2   0.8    -1   4.06
2 Doc2     welfare      1   1.8   0.2   0.8    -1   3.32
3 Doc3     welfare      2   2.2   0.2   0.8     1  24.5
4 Doc4     welfare      2   2     0.2   0.8     1  20.1
5 Doc1     tax          1   2     0.1   0.5    -1   4.95
6 Doc2     tax          1   1.8   0.1   0.5    -1   4.06
```

Wordshoal R Manual Example (Cont.)

```r
author_word <- wordshoal_grid %>%
  group_by(author, word) %>%
  summarise(lambda_total = sum(lambda),
            beta = first(beta),
            signal = lambda_total * beta, .groups="drop") # Calculate Total Lambda x Beta (Signal)

tibble(author_word)
```

```
# A tibble: 6 x 5
  author word     lambda_total  beta signal
   <dbl> <chr>           <dbl> <dbl>  <dbl>
1      1 military         5.46   1.2   6.55
2      1 tax              9.01   0.5   4.50
3      1 welfare          7.38   0.8   5.90
4      2 military        73.6    1.2  88.3
5      2 tax             29.9    0.5  15.0
6      2 welfare         44.6    0.8  35.7
```

Wordshoal R Manual Example (Cont.)

```r
tibble(author_word %>%
  group_by(author) %>%
  summarise(theta_signal = sum(signal))) # Theta Signal
```

```
# A tibble: 2 x 2
  author theta_signal
   <dbl>        <dbl>
1      1         17.0
2      2        139.
```

Wordshoal Quanteda (Irish Dail – Multiple Debates)

```r
library(wordshoal) # Load Wordshoal Package

data_corpus_irish30 <- wordshoal::data_corpus_irish30 # Speeches from 30th Irish Dail

names(docvars(data_corpus_irish30)) # Docvars
```

```
[1] "debateID"     "memberID"     "member.name"  "party.name"   "party.abbrev" "party.colour" "in.coaliti
```

```r
head(tibble(docvars(data_corpus_irish30))) # Sample
```

```
# A tibble: 6 x 7
  debateID memberID member.name          party.name  party.abbrev party.colour in.coalition
     <dbl>    <int> <chr>                <chr>       <chr>        <chr>        <lgl>
1       21        9 Mr. Michael Ahern    Fianna Fáil FF           #66BB66      TRUE
2       21       26 Mr. Sean Barrett     Fine Gael   FG           #6699FF      FALSE
3       21      106 Mr. Tommy Broughan   Labour      LAB          #CC0000      FALSE
4       21      119 Mr. Richard Bruton   Fine Gael   FG           #6699FF      FALSE
5       21      133 Ms. Joan Burton      Labour      LAB          #CC0000      FALSE
6       21      212 Mr. Paul Connaughton Fine Gael   FG           #6699FF      FALSE
```

```r
wordshoal_dfm <- dfm(tokens(data_corpus_irish30))
```

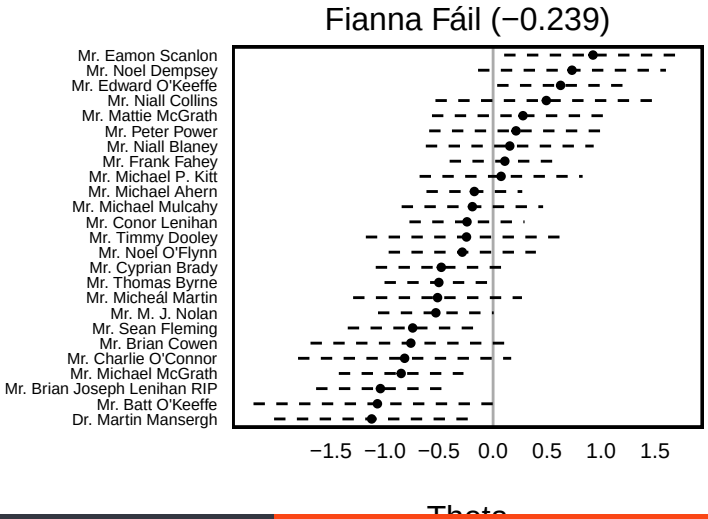
Wordshoal Quanteda (Cont.)

```
combined_thetas <- merge(as.data.frame(summary(wordshoal_fit)[[2]]),
                         docvars(data_corpus_irish30),
                         by.x = "row.names", by.y = "member.name") %>%
  distinct(memberID, .keep_all = T) # Combine Thetas & Associated Meta

head(tibble(combined_thetas)) # Print Head
```

```
# A tibble: 6 x 9
  Row.names               theta    se debateID memberID party.name  party.abbrev party.colour in.coalitio
  <I<chr>>                <dbl> <dbl>    <dbl>    <int> <chr>       <chr>        <chr>        <lgl>
1 Dr. Leo Varadkar        0.291  0.545       21     2192 Fine Gael   FG           #6699FF      FALSE
2 Dr. Martin Mansergh    -1.12   0.903       30     2002 Fianna Fáil FF           #66BB66      TRUE
3 Dr. Mary Upton          0.380  0.729       27     1751 Labour      LAB          #CC0000      FALSE
4 Mr. Aengus Ó Snodaigh   1.22   1.26        30     1829 Sinn Féin   SF           #008800      FALSE
5 Mr. Alan Shatter       -0.0665 0.507       21     1028 Fine Gael   FG           #6699FF      FALSE
6 Mr. Arthur Morgan       0.146  0.427       26     1824 Sinn Féin   SF           #008800      FALSE
```

```
aggregate_party_thetas <- aggregate(theta ~ party.name, data = combined_thetas, mean) %>%
  arrange(theta) # Aggregated by Party
```

Wordshoal Quanteda (Cont.)



Additional Notes Re: Wordshoal

- Intuitive – Scales well with uncertainty
- Capable of diagnosing both θ_i but β_j across documents (debates)
- Missingness = Random Error

Looking Forward (Next Class)

- We're done!
- I'll stick around until 6

Day	Student	Reviewer 1	Reviewer 2
4/12	Arora, Manya	Mamun	Revilla, Valeria
4/12	Cui, Lixiao	Meneley, Cory	Sarver, Sarah
4/12	Giron-Khalil, Jailyn	Mohammed, Suren	Sultana, Abida
4/12	Hao, Tianshu	Oneil, Connor	Tierman, Hunter
4/12	Islam, Md Sahidul	Revilla, Valeria	Trenchi, Alejandro
4/12	Mamun	Sarver, Sarah	Wang, Qianxi
4/12	Meneley, Cory	Sultana, Abida	Arora, Manya
4/12	Mohammed, Suren	Tierman, Hunter	Cui, Lixiao